MCA240102



PYTHON PROGRAMMING

UNIT 1 - Introduction

Assignment Questions:

1. Write a program to input the basic salary of an employee and calculate its Gross salary according to the following:

Basic Salary <= 10000 : HRA = 20%, DA = 80%
Basic Salary <= 20000 : HRA = 25%, DA = 90%
Basic Salary > 20000 : HRA = 30%, DA = 95%

Solution:

```
# Calculate Gross Salary
basic_salary = float(input("Enter basic salary: "))
if basic_salary <= 10000:
    hra = basic_salary * 20 / 100
    da = basic_salary * 80 / 100
elif basic_salary <= 20000:
    hra = basic_salary * 25 / 100
    da = basic_salary * 90 / 100
else:
    hra = basic_salary * 30 / 100
    da = basic_salary * 95 / 100
gross_salary = basic_salary + hra + da
print("HRA =", hra)
print("DA =", da)
print("Gross Salary =", gross_salary)
```

2. Write a menu driven python program which performs the following: (Implement using if-else)

a. Find area of circle (3.14 * r * r)
b. Find area of triangle (0.5 * base * height)
c. Find area of square and rectangle (side * side)(length * breath)
d. Find Simple Interest

Solution:

```
# Menu-driven program
print("----- MENU -----")
print("1. Area of Circle")
```

```
print("2. Area of Triangle")
print("3. Area of Square")
print("4. Area of Rectangle")
print("5. Simple Interest")
choice = int(input("Enter your choice: "))
if choice == 1:
    radius = float(input("Enter radius: "))
    area = 3.14 * radius * radius
    print("Area of Circle =", area)
elif choice == 2:
    base = float(input("Enter base: "))
    height = float(input("Enter height: "))
    area = 0.5 * base * height
    print("Area of Triangle =", area)
elif choice == 3:
    side = float(input("Enter side: "))
    area = side * side
    print("Area of Square =", area)
elif choice == 4:
    length = float(input("Enter length: "))
    breadth = float(input("Enter breadth: "))
    area = length * breadth
    print("Area of Rectangle =", area)
elif choice == 5:
    principal = float(input("Enter principal amount: "))
    rate = float(input("Enter rate of interest: "))
    time = float(input("Enter time in years: "))
    si = (principal * rate * time) / 100
    print("Simple Interest =", si)
else:
    print("Invalid choice")
```

3. Calculate Employee Bonus. Write a Python program to calculate an employee's bonus based on their basic salary and years of service:

Less than 2 years: 5% of basic salary
2 to 5 years: 10% of basic salary
6 to 10 years: 15% of basic salary
More than 10 years: 20% of basic salary

Calculate and display: Basic Salary , Bonus , Gross Salary = Basic Salary + Bonus

Solution:

```
# Calculate Employee Bonus
basic_salary = float(input("Enter basic salary: "))
years = int(input("Enter years of service: "))
if years < 2:
    bonus = basic_salary * 5 / 100
elif years <= 5:
    bonus = basic_salary * 10 / 100
elif years <= 10:
```

```
    bonus = basic_salary * 15 / 100
else:
    bonus = basic_salary * 20 / 100
gross_salary = basic_salary + bonus
print("Basic Salary =", basic_salary)
print("Bonus =", bonus)
print("Gross Salary =", gross_salary)
```

4. Create a list and perform the following methods: insert(), remove(), append(), len(), pop(), delete().

Solution:

```
# Create a list of computer languages
languages = ["Python", "Java", "C++", "JavaScript"]

print("Original List:", languages)

# 1. insert()
languages.insert(1, "C")
print("After insert():", languages)

# 2. remove()
languages.remove("C++")
print("After remove():", languages)

# 3. append()
languages.append("PHP")
print("After append():", languages)

# 4. len()
print("Length of list:", len(languages))

# 5. pop()
languages.pop()
print("After pop():", languages)

# 6. clear()
# languages.clear()
# print("After clear():", languages)

# 7. delete using del
del languages[1]
print("After delete (del):", languages)
```

5. Create a tuple and perform the following methods: Add items(), Check item in a tuple(), len(), Access items()

Solution:

```
# create a tuple
languages = ("C++","Java","PHP")
```

```
#  Add item in tuple
languages = ("React",) + languages + ("Java",)
print("After adding elements to tuple :",languages)

# check the item is available in tuple
if "React" in languages:
    print("React is in the tuple")
else:
    print("React is not in the language tuple")

# check the length of the tuple
print("Length of the tuple :",len(languages))

# Access the elements of the tuple
print("First item of the tuple :",languages[0])
print("First item of the tuple :",languages[1])
print("Last item of the tuple :",languages[-1])
```

6. Create a dictionary and apply the following methods:
Print the dictionary items, Access items, Use get(), Change values, Use len()

Solution:

```
# Create a dictionary
student = {
    "name": "Rahul",
    "course": "MCA",
    "semester": 3,
    "marks": 85
}
# 1. Print the dictionary items
print("Dictionary:", student)

# 2. Access items
print("Student Name:", student["name"])
print("Course:", student["course"])

# 3. Use get()
print("Semester:", student.get("semester"))
print("Marks:", student.get("marks"))

# 4. Change values
student["marks"] = 90
student["semester"] = 4

print("Dictionary after changing values:", student)

# 5. Use len()
print("Number of items:", len(student))
```

7. Write a program to create a menu with the following options:
Accept user input and perform the operations. Use function with arguments
To perform addition
To perform subtraction
To perform multiplication
To perform division

Solution:

```
def addition(a,b):
   return a + b
def subtraction(a,b):
   return a - b
def multiplication(a, b):
   return a * b
def division(a,b):
   return a / b

print("------------Menu----------")
print("1. Addition ")
print("2. Subtraction ")
print("3. Multiplication ")
print("4. Division ")
choice = int(input("Please select the choice (1-4): "))
num1 = int(input("Please enter Num1 : "))
num2 = int(input("Please enter Num2 : "))

if choice==1:
   print("Addition = ",addition(num1,num2))
elif choice==2:
   print("Subtraction = ",subtraction(num1,num2))
elif choice==3:
   print("Multiplication = ",multiplication(num1,num2))
elif choice==4:
   if (num2 != 0):
      print("Division =",division(num1, num2))
   else:
      print("Can not divide by zero")
else:
   print("Invalid Choice")
```

8. Write a Python program to input marks of the three subjects. Calculate the percentage and grade according to the following:

Percentage	>=	90% :	Grade	O
Percentage	>=	80% :	Grade	A
Percentage	>=	70% :	Grade	B
Percentage	>=	60% :	Grade	C
Percentage	>=	50% :	Grade	D
Percentage	>=	40% :	Grade	E
Percentage	<	40% :	Grade	F

Solution:

```
print("----- Percentage Calculations -----")
marks1 = int(input("Enter marks of subject1 : "))
marks2 = int(input("Enter marks of subject2 : "))
marks3 = int(input("Enter marks of subject3 : "))

total = marks1 + marks2 + marks3
percentage = total / 3

if percentage >= 90:
   grade = "O"
elif percentage >= 80:
   grade = "A"
elif percentage >= 70:
   grade = "B"
elif percentage >= 60:
   grade = "C"
elif percentage >= 50:
   grade = "D"
elif percentage >= 40:
   grade = "E"
elif percentage < 40:
   grade = "F"

print("="*25)
print("Subject1 = ",marks1)
print("Subject2 = ",marks2)
print("Subject3 = ",marks3)
print("Total = ",total)
print("Percentage = {:.2f}%".format(percentage))
print("Grade = ",grade)
print("="*25)
```

Practice Questions:

1. Write a program that evaluates an expression at run time using eval() function.

Solution :

```
# Accept an expression from the user
expression = input("Enter an expression: ")
# Evaluate the expression using eval()
result = eval(expression)
# Display the result
print("Result =", result)
```

2. Create a dictionary that accepts cricket players'names and scores in a match. Also we are retrieving runs by entering the player's name.

```
# Create an empty dictionary
players = {}
# Accept player names and scores
```

```
n = int(input("Enter number of players: "))

for i in range(n):
    name = input("Enter player name: ")
    score = int(input("Enter score: "))
    players[name] = score
# Display dictionary
print("\nPlayers and Scores:")
print(players)
# Retrieve runs by entering player's name
search_name = input("\nEnter player name to find score: ")

if search_name in players:
    print(search_name, "scored", players[search_name], "runs.")
else:
   print("Player not found.")
```

3. Write a python program to print number is positive/negative using if-else statement.

Solution:

```
num = int(input("Enter a number: "))
if num >= 0:
print("Number is Positive")
else:
print("Number is Negative")
```

4. Write a Python program to find the maximum between three numbers.

Solution:

```
a = int(input("Enter first number: "))
b = int(input("Enter second number: "))
c = int(input("Enter third number: "))

if a >= b and a >= c:
    maximum = a
elif b >= a and b >= c:
    maximum = b
else:
    maximum = c

print("Maximum number =", maximum)
```

5. Write a Python program that prompts users to enter numbers. The process will repeat until user enters. Finally, the program prints the sum of the numbers entered by the user.

Solution:

```
total = 0

while True:
    num = int(input("Enter a number (0 to stop): "))
```

```
    if num == 0:
        break

    total = total + num

print("Sum =", total)
```

6. Write a Python program to print all the numbers from 1 to 1000 that are not divisible by 2, 3, 5, 7, 11, 13, 17 and 19.

Solution:

```
for num in range(1, 1001):

    if (num % 2 != 0 and
        num % 3 != 0 and
        num % 5 != 0 and
        num % 7 != 0 and
        num % 11 != 0 and
        num % 13 != 0 and
        num % 17 != 0 and
        num % 19 != 0):

        print(num)
```

7. Write a Python program to check whether a number is an Armstrong number or not.

Solution:

```
# 153 = (1*3 + 5*3 + 3*3) = 153 where 3 is length
num = int(input("Enter a number: "))

temp = num
len = len(str(num))
total = 0

while temp > 0:
    digit = temp % 10
    #print(digit)
    total = total + digit ** len
    #print(total)
    temp = temp // 10
    #print(temp)
if total == num:
    print("Armstrong Number")
else:
    print("Not an Armstrong Number")
```

8. Write a python program for filter() to filter only even numbers from a given list.

Solution:

```
#Example: 6 = 1 + 2+ 3 check the factor
num = int(input("Enter a number: "))
total = 0
```

```
for i in range(1, num):
    if num % i == 0:
        total = total + i

if total == num:
    print("Perfect Number")
else:
    print("Not a Perfect Number")
```

9. Write a python program to print date, time for today and now.

Solution:

```
from datetime import datetime

now = datetime.now()

print("Today's Date:", now.date())
print("Current Time:", now.time())
print("Date and Time:", now)
```

10. Write a python program to add some days to your present date and print the date added

Solution:

```
from datetime import date, timedelta
# timedelta will handle the leap year and also the no of days available in months
days = int(input("Enter number of days to add: "))
today = date.today()
new_date = today + timedelta(days=days)

print("Today's date:", today)
print("Date after adding days:", new_date)
```

11. Write a Python program to print the sum of the series 1/2+1/3+1/4+ ... +1/N. Where N is a natural number.

Solution:

```
n = int(input("Enter N: "))
total = 0

for i in range(2, n + 1):
    total = total + (1 / i)

print("Sum of series =", total)
```

12. Write a python program to count frequency of characters in a given file check whether a number is perfect or not.

Solution:

```
file = open("sample.txt", "r")

text = file.read()
```

```
frequency = {}

for char in text:
    if char in frequency:
        frequency[char] = frequency[char] + 1
    else:
        frequency[char] = 1

file.close()

print("Character Frequency:")

for char, count in frequency.items():
    print(char, ":", count)
```

13. Write a Python program to find HCF (GCD) of two numbers.

Solution:

```
a = int(input("Enter first number: "))
b = int(input("Enter second number: "))

while b != 0:
    remainder = a % b
    a = b
    b = remainder

print("HCF / GCD =", a)
```

14. Write a Python program to swap first and last digits of a number.

```
num = int(input("Enter a number: "))

s = str(num)

if len(s) == 1:
    result = s
else:
    result = s[-1] + s[1:-1] + s[0]

print("Number after swapping:", result)
```

15. Write a Python program for printing prime numbers up to N. (N>100).

Solution:

```
n = int(input("Enter N: "))

print("Prime numbers:")

for num in range(2, n + 1):

    is_prime = True
```

```
    for i in range(2, num):
        if num % i == 0:
            is_prime = False
            break

    if is_prime:
        print(num, end=" ")
```

16. Write a program to find the sum of all odd numbers between 1 to n.

Solution:

```
n = int(input("Enter N: "))

total = 0

for i in range(1, n + 1, 2):
    total = total + i

print("Sum of odd numbers =", total)
```

17. Write a program to check whether a number is a palindrome or not.

Solution:

```
num = int(input("Enter a number: "))
original = num
reverse = 0

while num > 0:
    digit = num % 10
    reverse = reverse * 10 + digit
    num = num // 10

if original == reverse:
    print("Palindrome Number")
else:
    print("Not a Palindrome Number")
```

18. Write a python program to count the number of characters in the string and store them in a dictionary data structure

Solution:

```
text = input("Enter a string: ")
frequency = {}
for char in text:
    if char in frequency:
        frequency[char] = frequency[char] + 1
    else:
        frequency[char] = 1

print("Character frequency:")
print(frequency)
```

19. Write a program to print a multiplication table of any number.

Solution:

```
num = int(input("Enter a number: "))

for i in range(1, 11):
    print(num, "x", i, "=", num * i)
```

20. Write a function that will find power of a number.

```
base = int(input("Enter base: "))
power = int(input("Enter power: "))

result = base ** power
print("Result =", result)
```

21. Write a function that count number of digits in a number.

Solution:

```
def count_digits(num):
    count = 0

    while num > 0:
        count = count + 1
        num = num // 10

    return count

num = int(input("Enter a number: "))
result = count_digits(num)
print("Number of digits =", result)
```

22. find a sum of 15 + 25 +……+n5.

Solution:

23. Write a program to find the factorial value of a number using recursion.

Solution:

```
def factorial(n):
    if n == 0 or n == 1:
        return 1

    return n * factorial(n - 1)
num = int(input("Enter a number: "))
print("Factorial =", factorial(num))
```

24. Write a python program for printing Fibonacci series

Solution:

```
n = int(input("Enter number of terms: "))
a = 0
b = 1
print("Fibonacci Series:")

for i in range(n):
```

```
    print(a, end=" ")

    c = a + b
    a = b
    b = c
```

25. Write a function for finding factors (n) and use factors function to check whether a given number n is prime or not.

Solution:

```
def factors(n):
    result = []
    for i in range(1, n + 1):
        if n % i == 0:
            result.append(i)

    return result

num = int(input("Enter a number: "))
f = factors(num)
print("Factors:", f)
if len(f) == 2:
    print("Prime Number")
else:
    print("Not a Prime Number")
```